

Technologies and NuGet Packages

Technology baseline

Technology	Version/Policy	Purpose
.NET	8	Runtime and application platform
ASP.NET Core	8	Controller-based Web API
C#	12	Application language
YARP	2.3.0	Reverse proxy, routing, health, load balancing
Swashbuckle	10.2.3	Swagger/OpenAPI generation and UI
Microsoft.Data.SqlClient	7.0.2	MSSQL durable persistence
StackExchange.Redis	3.0.17	Transient Redis state
JwtBearer	8.0.29	Gateway/API JWT validation
xUnit v3	3.2.2	Unit-test framework
Moq	4.20.72	Application-boundary mocks

Package policy

Paid/freemium NuGet dependencies are not allowed. Selection order:

- 1 built-in .NET/ASP.NET Core capability;
- 2 an appropriate Microsoft-maintained package;
- 3 only fully free/open-source third-party packages with compatible licensing.

All versions are centrally pinned in Directory.Packages.props.

JwtBearer

Used for JWT validation at Gateway and FinWallet.Api. Hand-written JWT parsing/signing is avoided because it introduces unnecessary security risk. The signing algorithm remains a code-level invariant.

Microsoft.Data.SqlClient

Used for explicit async SQL, transactions, locking and financial persistence. MSSQL is the durable financial authority. Financial SQL values must be parameterized. Generic ORM/repository abstractions must not hide important locking behavior.

StackExchange.Redis

Used for OTP and other transient distributed state. ConnectionMultiplexer is singleton. Redis never becomes the Wallet/Ledger/idempotency financial source of truth.

Yarp.ReverseProxy

Used for Gateway route matching, authorization policies, transforms, cluster load balancing, health checks and transport tuning. It does not contain financial business rules.

Swashbuckle.AspNetCore

Provides common Swagger generation/UI across all Web APIs through FinWallet.Shared.Web. It is disabled by default in production.

xUnit v3 + Moq

Used for Application orchestration unit tests. Moq is appropriate for isolating external-provider/persistence boundary calls, but does not prove real MSSQL locking, Redis Lua or YARP routing behavior.

Framework-only capabilities

No separate package is added for:

- Kestrel limits;
- ASP.NET Core rate limiting;
- CORS;
- security-header middleware;
- HttpClientFactory/SocketsHttpHandler;
- PBKDF2/HMAC/SHA cryptography;
- CancellationToken/TimeProvider.

PBKDF2 note

PBKDF2 V1 work factor is not a loose appsettings tuning value. Existing credential rows do not store per-password iteration metadata, so changing it requires a versioned hash migration/rehash design.

New-package approval record

Every future package must document exact ID, version, license, owner project, purpose, why built-in functionality is insufficient, considered alternatives and financial/security impact.